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(54) **HAIR CONDITIONING COMPOSITION FOR IMPROVED DEPOSITION**

(71) Applicant: **Conopco, Inc.**, Englewood Cliffs, NJ (US)

(72) Inventors: **Richard Jonathan Barfoot**, Chesterfield (GB); **Michael James Cooke**, Wirral (GB); **Kelvin Brian Dickinson**, Wirral (GB); **Hailey Kelso**, Chester (GB); **Cesar Ernesto Mendoza Fernandez**, Liverpool (GB); **Paul Damien Price**, Wirral (GB); **Neil Scott Shaw**, Rochdale (GB)

(73) Assignee: **CONOPCO, INC.**, Englewood Cliffs, NJ (US)

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None

See application file for complete search history.

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Primary Examiner — Kyung S Chang

(74) *Attorney, Agent, or Firm* — Carter, DeLuca & Farrell, LLP; George Likourezos; Bret P. Shapiro

(57)

ABSTRACT

A composition comprising: (i) 0.01 to 10 wt % of a linear, cationic conditioning surfactant; (ii) 0.1 to 10 wt % of a linear fatty material; (iii) a particulate benefit agent selected from conditioning actives, scalp actives, encapsulated fragrance, emulsified fragrance, and mixtures thereof; (iv) 0.01 to 5 wt %, at 100% active, of a branched cationic co-surfactant, selected from structure 1, structure 2, structure 3 and mixtures thereof wherein: R₁, R₂, R₅ and R₆ comprise linear alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₄ to C₂₀, preferably from C₆ to C₁₈; R₃ and R₄ comprise linear or branched alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₆ to C₂₂; preferably from C₆ to C₁₂ n and m have a range of from 0 to 10, preferably selected from 0 and 1; p has a range of from 1 to 6, preferably selected from 1 and 2; R₇ comprises an alkyl chain having a carbon-carbon chain length of from C₁ to C₄, preferably C₁ to C₂; R₈ comprises a proton or an alkyl chain having a carbon-carbon chain length of from C₁ to C₄, preferably C₁ to C₂; and X is an organic or inorganic anion; wherein the molar ratios of branched cationic co-surfactants (iv) to linear cationic surfactants (i) are in the range of from 1:20 to 1:1. results in improved particulate benefit agent deposition onto hair.

17 Claims, No Drawings

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HAIR CONDITIONING COMPOSITION FOR IMPROVED DEPOSITION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Stage Application under 35 U.S.C. § 371 of International Application No. PCT/EP2019/084382, filed on Dec. 10, 2019, which claims the benefit of European Patent Application No. 18214094.7, filed Dec. 19, 2018, the entire disclosures of which are hereby incorporated by reference for any and all purposes.

FIELD OF THE INVENTION

The invention is concerned with conditioning compositions, containing a branched co-surfactant, for the treatment of hair, which comprise a benefit agent, for example silicone, to be deposited onto the hair during use and particularly relates to a conditioning composition that enables increased amounts of benefit agent to be deposited.

BACKGROUND AND PRIOR ART

In personal care compositions, such as hair treatment compositions, the deposition and delivery of benefit agents are often key drivers of product performance. For example, many of the hair conditioner products in the market today work to deliver benefits to hair by depositing benefit agents such as fragrance materials, silicones and damage repair actives onto the hair during the wash and care process.

However, consumers report being disappointed by the level of benefit derived from use of some compositions. This is usually caused by insufficient amount of benefit agents being delivered to the surface. It is, therefore, desirable to develop compositions that provide improved delivery of benefit materials to a surface, for example hair.

Various types of branched cationic compounds are known in hair treatment compositions for a variety of benefits.

WO 17/172117 discloses a composition for treating keratinous substrates comprising a cationic agent comprising a defined first quaternary ammonium compound and an imidazoline compound, a modified starch, two silane compounds, a cationic vinylpyrrolidone polymer and water. Hair treated with the compositions is purported to have improved mass, body, volume, to be easily rinsed, to dry fast, to stay clean longer and be sufficiently conditioned. US 2005/175569 discloses cosmetic compositions, for example for conditioning and styling hair, comprising a cationic surfactant, which may be a quaternary ammonium salt.

JP 2005-060271 discloses an aqueous hair cosmetic composition that can comprise (A) a dimethylpolysiloxane represented by general formula (1), (B) a dimethylpolysiloxane represented by general formula (2), (C) a cyclic dimethylpolysiloxane represented by general formula (3) at a ratio of $[(B)+(C)]/(A)$ greater than or equal to 1; and (D) an additional quaternary ammonium component. The composition is said to provide a range of conditioning benefits to hair in the wet, rinse and dry stages.

Our own published applications WO 02/102334 and WO 01/43718 provide aqueous hair treatment compositions having cleansing and conditioning properties that comprise quaternary ammonium based cationic surfactants having defined hydrocarbyl chains.

Glycine betaine derivatives are known in home and personal care products. These derivatives have been used in hair applications to deliver conditioning benefits to hair.

US 2017/087074 (Centre National De La Recherche Scientifique) discloses surfactant compositions for use in cosmetic and pharmaceutical compositions. The compositions include an alkylpolyglucoside grafted with glycine betaine in combination with a glycine betaine and are stable. Similar compositions are disclosed in US 2017/087077.

US 2014/246041 (Henkel) discloses a composition for conditioning hair containing a defined esterquat which has a structure that can cover glycine betaine derivatives and at least one care-providing substance(s) selected from the group of L-carnitine and/or salts thereof, and/or taurine and/or salts thereof, and/or vitamins and vitamin precursors, and/or niacinamide, and/or ubiquinone, and/or ectoin. WO 13/083349 discloses a similar esterquat for conditioning keratin fibres.

EP 1 016650 (Kao) disclose a method of preparation of a betaine alkyl ester which can be used in hair cosmetics to give feel benefits to hair. Cetanol (ex Kao Corporation) is used in the formulation examples.

U.S. Pat. No. 5,374,421 discloses a composition for hair treatment containing (a) 0.1-10 wt. % of a modified silicone polymer having at least one alk-oxy group in the molecule and a melting point of not lower than 30° C., (b) 0.1-20 wt. % of a cationic surface active agent, (c) 0.1-30 wt. % of an oily or fatty material, (d) 0.1-90 wt. % of an organic liquid which is compatible with water and of which molecule has at least one hydroxy group, and (e) water. Cetostearyltrimethyl ammonium chloride is exemplified as a cationic conditioning material and myristyloxy modified silicone as particulate benefit agent.

Whilst branched materials are known in home and personal care products, they have not been applied effectively to provide improved deposition of benefit agents onto hair.

Product rheology is a key attribute to consumers. We have, however, found that adding branched surfactant materials into gel networks disrupts the gel bilayers and consequently reduces viscosity and yield stress to unacceptably low levels.

Despite the prior art, there remains a need to deliver improved delivery of benefits to hair without compromising on consumer desired viscosity characteristics.

We have now surprisingly found that compositions comprising a combination of certain branched co-surfactants in combination with defined linear conditioning surfactant provide an unexpectedly large enhancement in the deposition of benefit agents (eg silicone, encapsulated fragrances) whilst maintaining excellent product rheology.

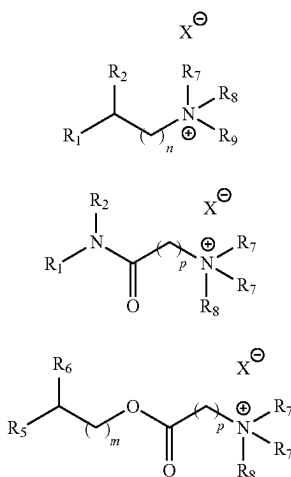
All percentages quoted herein are by weight based on total weight, unless otherwise stated.

DEFINITION OF THE INVENTION

Accordingly, there is provided a composition comprising:

- (i) 0.01 to 10 wt % of a linear, cationic conditioning surfactant;
- (ii) 0.1 to 10 wt % of a linear fatty material;
- (iii) a particulate benefit agent selected from conditioning actives, scalp actives, encapsulated fragrance, emulsified fragrance and mixtures thereof;
- (iv) 0.01 to 5 wt %, at 100% active, of a branched cationic co-surfactant, selected from structure 1, structure 2, structure 3 and mixtures thereof

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wherein:

R₁, R₂, R₅ and R₆ comprise linear alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₄ to C₂₀, preferably from C₆ to C₁₃;

R₃ and R₄ comprise linear or branched alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₆ to C₂₂; preferably from C₆ to C₁₂; n and m have a range of from 0 to 10, preferably selected from 0 and 1;

p has a range of from 1 to 6, preferably selected from 1 and 2;

R₇ comprises an alkyl chain having a carbon-carbon chain length of from C₁ to C₁₈, preferably C₁ to C₄; most preferably from C₁ to C₂

R₈ comprises a proton or an alkyl chain having a carbon-carbon chain length of from C₁ to C₁₈, preferably from C₁ to C₄, most preferably C₁ to C₂; and

X is an organic or inorganic anion;

wherein the molar ratios of branched cationic co-surfactants (iv) to linear cationic surfactants (i) are in the range of from 1:20 to 1:1, preferably from 1:10 to 1:1, most preferably 1:5 to 1:2.

In a second aspect, the invention provides a method of increasing deposition of a particulate benefit agent selected from conditioning actives, scalp actives, encapsulated fragrance, emulsified fragrance, and mixtures thereof to hair comprising the step of applying to hair a composition of the first aspect.

The method of the invention preferably comprises an additional step of rinsing the composition from the hair.

Preferably, the method is a method of increasing silicone deposition to hair comprising the steps of applying to hair a composition as defined by the first aspect of the invention and rinsing the hair with water.

Compositions in accordance with the invention are preferably formulated as conditioners for the treatment of hair (typically after shampooing) and subsequent rinsing.

GENERAL DESCRIPTION OF THE INVENTION

Preferably, the treatment composition is selected from a rinse-off hair conditioner, a hair mask, a leave-on conditioner composition, and a pre-treatment composition, more preferably selected from a rinse-off hair conditioner, a hair mask, a leave-on conditioner composition, and a pre-treat-

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Structure 1

ment composition, for example an oil treatment, and most preferably selected from a rinse-off hair conditioner, a hair mask and a leave-on conditioner composition. The treatment composition is preferably selected from a rinse-off hair conditioner and a leave-on conditioner.

Rinse off conditioners for use in the invention are conditioners that are typically left on wet hair for 1 to 2 minutes before being rinsed off.

Structure 2

Hair masks for use in the present invention are treatments that are typically left on the hair for 3 to 10 minutes, preferably from 3 to 5 minutes, more preferably 4 to 5 minutes, before being rinsed off.

Structure 3

Leave-on conditioners for use in the invention are typically applied to the hair and left on the hair for more than 10 minutes, and preferably are applied to the hair after washing and not rinsed out until the next wash.

The Linear Cationic Conditioning Surfactant

Conditioner compositions will comprise a linear cationic conditioning surfactant, which is cosmetically acceptable and suitable for topical application to the hair.

Preferably, the linear cationic conditioning surfactants have the formula 1: N⁺(R¹)(R²)(R³)(R⁴), wherein R¹, R², R³ and R⁴ are independently (C₁ to C₃₀) alkyl or benzyl.

In formula 1, preferably, one, two or three of R¹, R², R³ and R⁴ are independently (C₄ to C₃₀) alkyl and the other R¹, R², R³ and R⁴ group or groups are (C₁-C₆) alkyl or benzyl.

More preferably, one or two of R¹, R², R³ and R⁴ are independently (C₆ to C₃₀) alkyl and the other R¹, R², R³ and R⁴ groups are (C₁-C₆) alkyl or benzyl groups. Optionally, the alkyl groups may comprise one or more ester (—OCO— or —COO—), amido (—NOC— or NCO—), and/or ether (—O—) linkages within the alkyl chain. Alkyl groups may optionally be substituted with one or more hydroxyl groups. Alkyl groups may be straight chain or branched and, for alkyl groups having 3 or more carbon atoms, cyclic. The alkyl groups may be saturated or may contain one or more carbon-carbon double bonds (e.g., oleyl). Alkyl groups are optionally ethoxylated on the alkyl chain with one or more ethyleneoxy groups.

Suitable quaternary amine salts for use in conditioner compositions according to the invention are quaternary amine salt comprising from 12 to 24 carbon atoms, preferably from 16 to 22 carbon atoms.

Suitable quaternary amine salts for use in conditioner compositions according to the invention include cetyltrimethylammonium chloride, behenyltrimethylammonium chloride, Behentrimonium methosulphate, BehenylAmido Propyl Di-Methyl Amine, cetyltrimethylammonium chloride, cetylpyridinium chloride, tetramethylammonium chloride, tetraethylammonium chloride, octyltrimethylammonium chloride, dodecyltrimethylammonium chloride, hexadecyltrimethylammonium chloride, octyldimethylbenzylammonium chloride, decyldimethylbenzylammonium chloride, stearyldimethylbenzylammonium chloride, Stearalkonium Chloride, Stearalkonium methosulphate, didodecyltrimethylammonium chloride, dioctadecyldimethylammonium chloride, tallowtrimethylammonium chloride, dihydrogenated tallow dimethyl ammonium chloride (e.g., Arquad 2HT/75 from Akzo Nobel) and cocotrimethylammonium chloride.

Preferred quaternary amine salts selected from behenyltrimethylammonium chloride, Behentrimonium methosulphate, cetyltrimethylammonium chloride, and mixtures thereof.

A particularly useful cationic surfactant for use in conditioners according to the invention is cetyltrimethylammonium chloride, available commercially, for example as

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GENAMIN CTAC, ex Hoechst Celanese. Another particularly preferred cationic surfactant for use in conditioners according to the invention is behenyltrimethylammonium chloride, available commercially, for example as GENAMIN KDMP, ex Clariant.

Further suitable cationic surfactants include those materials having the CTFA designations Quaternium-5, Quaternium-31, and Quaternium-18. Mixtures of any of the foregoing materials may also be suitable.

Another example of a class of suitable cationic surfactants for use in the invention, either alone or together with one or more other cationic surfactants, is a combination of (i) and (ii) below:

- (i) an amidoamine corresponding to the general formula (II):



in which R^1 is a hydrocarbyl chain having 10 or more carbon atoms, R^2 and R^3 are independently selected from hydrocarbyl chains of from 1 to 10 carbon atoms, and m is an integer from 1 to about 10; and

- (ii) an acid.

As used herein, the term hydrocarbyl chain means an alkyl or alkenyl chain.

Preferred amidoamine compounds are those corresponding to formula (I) in which

R^1 is a hydrocarbyl residue having from about 11 to about 24 carbon atoms,

R^2 and R^3 are each independently hydrocarbyl residues, preferably alkyl groups, having from 1 to about 4 carbon atoms, and m is an integer from 1 to about 4.

Preferably, R^2 and R^3 are methyl or ethyl groups.

Preferably, m is 2 or 3, i.e. an ethylene or propylene group.

Preferred amidoamines useful herein include stearamidopropyl dimethylamine, stearamidopropyl diethylamine, stearamidoethyl diethylamine, stearamidoethyl dimethylamine, palmitamidopropyl dimethylamine, palmitamidopropyl diethylamine, palmitamidoethyl diethylamine, palmitamidoethyl dimethylamine, behenamidopropyl dimethylamine, behenamidopropyl diethylamine, behenamidoethyl diethylamine, behenamidoethyl dimethylamine, arachidamidopropyl dimethylamine, arachidamidopropyl diethylamine, arachidamidoethyl diethylamine, arachidamidoethyl dimethylamine, and mixtures thereof.

Particularly preferred amidoamines useful herein are stearamidopropyl dimethylamine, stearamidoethyl diethylamine, and mixtures thereof.

Commercially available amidoamines useful herein include: stearamidopropyl dimethylamine with tradenames LEXAMINE S-13 available from Inolex (Philadelphia Pennsylvania, USA) and AMIDOAMINE MSP available from Nikko (Tokyo, Japan), stearamidoethyl diethylamine with a tradename AMIDOAMINE S available from Nikko, behenamidopropyl dimethylamine with a tradename INCROMINE BB available from Croda (North Humberston, England), and various amidoamines with tradenames SCHERCODINE series available from Scher (Clifton New Jersey, USA).

Acid may be any organic or mineral acid which is capable of protonating the amidoamine in the conditioner composition. Suitable acids useful herein include hydrochloric acid, acetic acid, tartaric acid, fumaric acid, lactic acid, malic acid, succinic acid, and mixtures thereof. Preferably, the acid is selected from the group consisting of acetic acid, tartaric acid, hydrochloric acid, fumaric acid, lactic acid and mixtures thereof.

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The primary role of the acid is to protonate the amidoamine in the hair treatment composition thus forming a tertiary amine salt (TAS) in situ in the hair treatment composition. The TAS in effect is a non-permanent quaternary ammonium or pseudo-quaternary ammonium cationic surfactant.

Suitably, the acid is included in a sufficient amount to protonate more than 95 mole % (293 K) of the amidoamine present.

In conditioners for use in the invention, the level of linear cationic conditioning surfactant will generally range from 0.01 to 10%, more preferably 0.05 to 7.5%, most preferably 0.1 to 5% by total weight of cationic conditioning surfactant based on the total weight of the composition.

The Linear Fatty Material

The composition of the invention comprises from 0.1 to 10 wt % of a linear fatty material.

The combined use of fatty materials and cationic surfactants in conditioning compositions is believed to be especially advantageous, because this leads to the formation of a structured lamellar or liquid crystal phase, in which the cationic surfactant is dispersed.

By "fatty material" is meant a fatty alcohol, an alkoxyated fatty alcohol, a fatty acid or a mixture thereof. Preferably the linear fatty material is selected from a fatty alcohol and a fatty acid, most preferably a fatty alcohol.

Preferably, the alkyl chain of the fatty material is fully saturated. Representative fatty materials comprise from 8 to 22 carbon atoms, more preferably 16 to 22.

Suitable fatty alcohols comprise from 8 to 22 carbon atoms, preferably 16 to 22, most preferably C16 to C18. Fatty alcohols are typically compounds containing straight chain alkyl groups. Preferably, the alkyl groups are saturated. Examples of preferred fatty alcohols include cetyl alcohol, stearyl alcohol and mixtures thereof. The use of these materials is also advantageous in that they contribute to the overall conditioning properties of compositions for use in the invention.

Alkoxyated, (e.g. ethoxyated or propoxyated) fatty alcohols having from about 12 to about 18 carbon atoms in the alkyl chain can be used in place of, or in addition to, the fatty alcohols themselves. Suitable examples include ethylene glycol cetyl ether, polyoxyethylene (2) stearyl ether, polyoxyethylene (4) cetyl ether, and mixtures thereof.

The level of fatty material in conditioners of the invention is suitably from 0.01 to 10, preferably from 0.1 to 10, and more preferably from 0.1 to 5 percent by weight of the total composition. The weight ratio of cationic surfactant to fatty alcohol is suitably from 10:1 to 1:10, preferably from 4:1 to 1:8, optimally from 1:1 to 1:7, for example 1:3.

The Particulate Benefit Agent

The composition of the invention comprises a particulate benefit agent. The particulate benefit agent is selected from conditioning actives, scalp actives, encapsulated fragrance, emulsified fragrance, and mixtures thereof. More preferably the particulate benefit agent is selected from conditioning actives, encapsulated fragrance and mixtures thereof. Most preferably, the particulate benefit agent is selected from a silicone emulsion and an encapsulated fragrance.

Preferred conditioning actives are silicone emulsions.

Preferred silicone emulsions do not comprise a hydrophobic modification, preferably the silicone emulsion is not a myristyloxyl modified silicone, most preferably not a myristyloxyl modified silicone or a cetyloxyl modified silicone. Most preferably, the silicone emulsions for use in the

compositions of the invention are selected from emulsions of dimethicone, dimethiconol, amodimethicone and mixtures thereof.

The particulate benefit agent may be a scalp active, which is insoluble in the composition of the invention, or in the form of an emulsion. Preferred scalp actives are selected from metal pyrithiones, azoles, octopirox (piroctone olamine), selenium sulfide, salicylic acid and combinations thereof, preferably metal pyrithiones, azoles and octopirox. Azole based antifungal agents include ketoconazole and climbazole, preferably climbazole. The particulate benefit agent may be an emulsified fragrance or an encapsulated fragrance. For the sake of clarity, "fragrance" may also be referred to herein as "perfume". The following are perfume materials that may suitably be emulsified or encapsulated for use in the compositions of the invention.

Examples of perfume materials for use in the invention include geraniol, geranyl acetate, linalol, linalyl acetate, tetrahydrolinalol, citronellol, citronellyl acetate, dihydromyrcenol, dihydromyrcenyl acetate, tetrahydromyrcenol, terpineol, terpinyl acetate, nopyl acetate, 2-phenyl-ethanol, 2-pentylethyl acetate, benzyl alcohol, benzyl acetate, benzyl salicylate, styrallyl acetate, benzyl benzoate, amyl salicylate, dimethylbenzyl-carbinol, trichloromethylphenyl-carbinyl acetate, p-tert-butylcyclohexyl acetate, isononyl acetate, vetiveryl acetate, vetiverol, α -hexylcinnamaldehyde, 2-methyl-3-p-tert-butylphey)propanal, 2-methyl-3-(p-isopropylphenyl)propanal, 2-(p-tert-butylphey)propanal, 2,4-dimethyl-cyclohex-3-enyl-carboxaldehyde, tricyclodecenyl acetate, tricyclodecenyl propionate, 4-(4-hydroxy-4-methyl-pentyl)-3-cyclohexenecarboxaldehyde, 4-(4-methyl-3-pentenyl)-3-cyclohexenecarboxaldehyde, 4-acetoxy-3-pentyl-tetrahydropyran, 3-carboxymethyl-2-pentylcyclopentane, 2-n-heptylcyclopentanone, 3-methyl-2-pentyl-2-cyclopentenone, n-decanal, n-dodecanal, 9-decenol-1, phenoxyethyl isobutyrate, phenyl-acetaldehyde dimethyl-acetal, phenylacetaldehyde diethylacetal, geranyl nitrile, citronellyl nitrile, cedryl acetate, 3-isocamphylcyclohexanol, cedryl methyl ether, isolongifolanone, aubepine nitrile, aubepine, heliotropin, coumarin, eugenol, vanillin, diphenyl oxide, hydroxycitronellal, ionones, methylionones, isomethylionones, irones, cis-3-hexenol and esters thereof, indan musks, tetralin musks, isochroman musks, macrocyclic ketones, macrolactone musks, ethylene brassylate and mixtures thereof.

Encapsulated fragrances preferably comprise a polymeric shell (capsule wall) that forms a microcapsule. The polymeric shell of the microcapsule may be prepared using interfacial polymerisation.

Interfacial polymerisation produces encapsulated shells from the reaction of at least one oil-soluble wall forming material present in the oil phase with at least one water-soluble wall forming material present in the aqueous phase. A polymerisation reaction between the two wall-forming materials occurs resulting in the formation of covalent bonds at the interface of the oil and aqueous phases to form the capsule wall.

Preferably the polymeric shell of the microcapsule is an aminoplast resin selected from polyurea formed by reaction of polyisocyanates with material selected from polyamines, polyimines or mixtures thereof.

Preferably, the microcapsules are activated by shear; that is to say they are broken by shear to release the contents.

A particularly preferred microcapsule has a polyurea shell, prepared as described in US2013/0330292 A1 and US2012/0148644 A1 and available from International Flavors & Fragrances Inc.

Advantageously the polymeric shell comprises at most 20 wt % of the weight of the microcapsules.

By modifying process conditions microcapsules of a desired size can be produced in known manner. The microcapsules typically have a mean diameter in the range 1 to 500 microns, preferably 1 to 300 microns, more preferably 1 to 50 microns and most preferably 1 to 10 microns. If necessary, the microcapsules as initially produced may be filtered or screened to produce a product of greater size uniformity.

In a typical composition according to the invention the level of microcapsules will generally range from 0.2 to 2%, and preferably ranges from 0.5 to 1.5% by weight based on the total weight of the composition.

Silicone

The compositions of the invention can contain emulsified droplets of a silicone conditioning agent, which is preferably not hydrophobically modified.

Suitable silicones include polydimethylsiloxanes which have the CTFA designation dimethicone. Also suitable for use compositions of the invention are polydimethyl siloxanes having hydroxyl end groups, which have the CTFA designation dimethiconol. Preferably, the silicone is selected from the group consisting of dimethicone, dimethiconol, amodimethicone and mixtures thereof. Also preferred are blends of amino-functionalised silicones with dimethicones.

The viscosity of the emulsified silicone itself (not the emulsion or the final hair conditioning composition) is typically at least 10,000 cst at 25° C. the viscosity of the silicone itself is preferably at least 60,000 cst, most preferably at least 500,000 cst, ideally at least 1,000,000 cst. Preferably the viscosity does not exceed 109 cst for ease of formulation.

Emulsified silicones for use in the compositions of the invention will typically have a D90 silicone droplet size in the composition of less than 30, preferably less than 20, more preferably less than 10 micron, ideally from 0.01 to 1 micron. Silicone emulsions having an average silicone droplet size (D50) of 0.15 micron are generally termed micro-emulsions.

Silicone particle size may be measured by means of a laser light scattering technique, for example using a 2600D Particle Sizer from Malvern Instruments.

Examples of suitable pre-formed emulsions include Xiameter MEM 1785 and microemulsion DC2-1865 available from Dow Corning. These are emulsions/microemulsions of dimethiconol. Cross-linked silicone gums are also available in a pre-emulsified form, which is advantageous for ease of formulation.

A further preferred class of silicones for inclusion in compositions of the invention are amino functional silicones. By "amino functional silicone" is meant a silicone containing at least one primary, secondary or tertiary amine group, or a quaternary ammonium group. Examples of suitable amino functional silicones include: polysiloxanes having the CTFA designation "amodimethicone". A preferred amodimethicone is commercially available from Dow Corning as DC 7134.

Specific examples of amino functional silicones suitable for use in the invention are the aminosilicone oils DC2-8220, DC2-8166 and DC2-8566 (all ex Dow Corning).

Suitable quaternary silicone polymers are described in EP-A-0 530 974. A preferred quaternary silicone polymer is K3474, ex Goldschmidt.

Also suitable are emulsions of amino functional silicone oils with non ionic and/or cationic surfactant.

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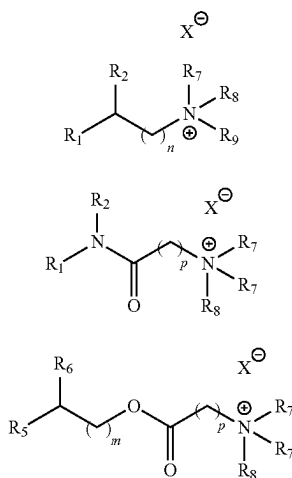
Pre-formed emulsions of amino functional silicone are also available from suppliers of silicone oils such as Dow Corning and General Electric. Specific examples include DC939 Cationic Emulsion and the non-ionic emulsions DC2-7224, DC2-8467, DC2-8177 and DC2-8154 (all ex

The total amount of silicone is preferably from 0.1 wt % to 10 wt % of the total composition more preferably from 0.1 wt % to 5 wt %, most preferably 0.25 wt % to 3 wt % is a suitable level.

The Branched Cationic Co-Surfactant

The composition of the invention comprises a branched cationic co-surfactant.

The branched cationic co-surfactant is selected from structure 1, structure 2, structure 3 and mixtures thereof.



wherein:

R₁, R₂, R₅ and R₆ comprise linear alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₄ to C₂₀, preferably from C₆ to C₁₈; optionally, at least one of R₁, R₂, R₅ and R₆ comprise a group selected from an ester group, and amido group and an ether group;

R₃ and R₄ comprise linear or branched alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₆ to C₂₂; preferably from C₆ to C₁₂; optionally, at least one of R₃ and R₄ comprise ester or amido groups;

n and m have a range of from 0 to 10, preferably selected from 0 and 1;

p has a range of from 1 to 6;

R₇ comprises an alkyl chain having a carbon-carbon chain length of from C₁ to C₄, preferably C₁ to C₂;

R₈ comprises a proton or an alkyl chain having a carbon-carbon chain length of from C₁ to C₄, preferably C₁ to C₂; and

X is an organic or inorganic anion;

The molar ratios of branched cationic co-surfactants (iv) to linear cationic surfactants (i) are in the range of from 1:20 to 1:1, preferably from 1:10 to 1:1, most preferably 1:5 to 1:2.

The variable, p has a range of from 1 to 6, preferably selected from 1 and 2, most preferably 1.

In structures 1-3, the amine head group is charged within the final formulation. Raw materials include, however, spe-

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cies where the charge is not permanent and can be induced by protonation in the formulation using a strong acid. When R₈ is a proton in the above general formulae therefore, the proton may be present in the raw material or become associated during formulation.

Optionally, at least one of R₁, R₂, R₃, R₄, R₅ and R₆ comprise linkages within the alkyl chain selected from the group consisting of an ester group (—OCO— or —COO—), an amido group (—NOC— or NCO—), and an ether group (—O—).

The branched co-surfactant is present in an amount of from 0.01 to 5 wt %, preferably 0.1 to 2, more preferably 0.1 to 1.0, most preferably 0.2 to 0.7 wt % (at 100% active and based on weight of total composition).

X is an organic or inorganic anion. Preferably, X comprises an anion selected from the halide ions; sulphates of the general formula RSO₃⁻, wherein R is a saturated or unsaturated alkyl radical having 1 to 4 carbon atoms, and anionic radicals of organic acids.

Preferred halide ions are selected from fluoride, chloride, bromide and iodide. Preferred anionic radicals of organic acids are selected from maleate, fumarate, oxalate, tartrate, citrate, lactate and acetate. Preferred sulphates are methanesulphonate and ethanesulphonate.

Most preferably, X⁻ comprises an anion selected from a halide, a methanesulphonate group and an ethanesulphonate group.

In a preferred embodiment,

R₁, R₂, R₅ and R₆ comprise linear alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₆ to C₁₈;

R₃ and R₄ comprise linear or branched alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₆ to C₁₂;

n and m are selected from 0 and 1;

p is 1;

R₇ comprises an alkyl chain having a carbon-carbon chain length of from C₁ to C₂;

R₈ comprises a proton or an alkyl chain having a carbon-carbon chain length of from C₁ to C₂; and

X is selected from a halide, methanesulphonate and ethanesulphonate.

An example of a suitable material specific to structure 1 is N,N,N-trimethyl-2-octyldodecan-1-aminium methanesulphonate.

An example of a suitable material specific to structure 2 is 2-(dioctylamino)-N,N,N-trimethyl-2-oxoethan-1-aminium methanesulphonate.

Examples of suitable materials conforming to structure 3 are 2-((2-butyloctyl)oxy)-N,N,N-trimethyl-2-oxoethan-1-aminium methanesulphonate, 2-((2-hexyldecyl)oxy)-N,N,N-trimethyl-2-oxoethan-1-aminium methanesulphonate, N,N,N-trimethyl-2-((2-octyldodecyl)oxy)-2-oxoethan-1-aminium methanesulphonate, 2-((2-decyltetradecyl)oxy)-N,N,N-trimethyl-2-oxoethan-1-aminium methanesulphonate, 2-((2-dodecylhexadecyl)oxy)-N,N,N-trimethyl-2-oxoethan-1-aminium methanesulphonate and N,N,N-trimethyl-2-oxo-2-((2-tetradecyloctadecyl)oxy)ethan-1-aminium methanesulphonate.

Composition Rheology

The compositions of the invention provide good viscosity and yield stress properties.

The compositions have a preferred yield stress range of from 30 to 200 Pascals (Pa), most preferably from 40 to 150 Pa peak value at 25° C. and 1 Hz. The method to measure the yield stress uses a serrated parallel-plate geometry, 40 mm in diameter, attached to a suitable rheometer capable of

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applying oscillations at a constant frequency of 1 Hz, and an amplitude sweep in the range of 0.1% to 2000%. The amplitude sweep range is applied at no more than ten points per decade of strain range covered at no more than 4 cycles per amplitude. The instrument should be operated under controlled strain, such as with the ARES G2 Rheometer from TA Instruments. The geometry's temperature should be set at 25° C. by means of, for example, a Peltier-controlled plate, or a recirculating bath. The yield stress is determined by plotting the elastic stress against strain amplitude, and at the peak of the curve, the maximum value is quoted as the yield stress. The elastic stress is calculated as the multiplication of (storage modulus)*(strain amplitude), each readily obtained from the instrument.

The compositions have a viscosity of from 5,000 to 750,000 centipoise, preferably from 50,000 to 600,000 centipoise, more preferably from 50,000 to 450,000 as measured at 30° C. on a Brookfield RVT using a Spindle A or B at 0.5 rpm for 60 seconds on a Helipath stand.

A preferred conditioner comprises a conditioning gel phase. These conditioners have little or no vesicle content. Such conditioners and methods for making them are described in WO2014/016354, WO2014/016353, WO2012/016352 and WO2014/016351.

Such a conditioning gel phase comprises, by total weight of the composition,

i) from 0.4 to 8 wt % of fatty alcohol having from 8 to 22 carbons,

ii) from 0.1 to 2 wt % of cationic surfactant,

and the composition confers a Draw Mass of from 1 to 250 g, preferably 2 to 100 g, more preferably 2 to 50 g, even more preferably 5 to 40 g and most preferably 5 to 25 g to hair treated with the composition.

Draw Mass is the mass required to draw a hair switch through a comb or brush. Thus the more tangled the hair the greater the mass required to pull the switch through the comb or brush, and the greater the level of condition of the hair, the lower the Draw Mass.

The Draw Mass is the mass required to draw a hair switch, for example of weight 1 to 20 g, length 10 to 30 cm, and width 0.5 to 5 cm through a comb or brush, as measured by first placing the hair switch onto the comb or brush, such that from 5 to 20 cm of hair is left hanging at the glued end of the switch, and then adding weights to the hanging end until the switch falls through the comb or brush.

Preferably, the hair switch is of weight 1 to 20 g, more preferably 2 to 15 g, most preferably from 5 to 10 g. Preferably, the hair switch has a length of from 10 to 40 cm, more preferably from 10 to 30 cm, and a width of from 0.5 to 5 cm, more preferably from 1.5 to 4 cm.

Most preferably, the Draw Mass is the mass required to draw a hair switch, for example of weight 10 g, length 20 cm, and width 3 cm through a comb or brush, as measured by first placing the hair switch onto the comb or brush, such that from 20 cm of hair is left hanging at the glued end of the switch, and then adding weights to the hanging end until the switch falls through the comb or brush.

Further Ingredients

The composition according to the invention may comprise any of a number of ingredients which are common to hair conditioning compositions.

Other ingredients may include, preservatives, colouring agents, polyols such as glycerine and polypropylene glycol, chelating agents such as EDTA, antioxidants such as vitamin E acetate, fragrances, antimicrobials and sunscreens. Each of these ingredients will be present in an amount effective to accomplish its purpose. Generally these optional ingredients

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are included individually at a level of up to about 5% by weight of the total composition.

Preferably, the further ingredients include perfumes, preservatives, colours and conditioning silicones.

The compositions of the invention are preferably free from viscosity modifiers and thickening agents for example thickening polymers.

Mixtures of any of the above active ingredients may also be used.

Generally, such ingredients are included individually at a level of up to 2%, preferably up to 1%, by weight of the total composition.

Embodiments of the invention are given in the following examples, in which all percentages are quoted by weight based on total weight unless otherwise stated.

EXAMPLES

Example 1: Compositions 1-4 in Accordance with the Invention and Comparative Composition A

The following compositions were prepared:

TABLE 1

Compositions of example A (comparative) and examples 1 to 3 (in accordance with the invention).				
Ingredients	Example A Comparative	Example 1 Structure 1	Example 2 Structure 2	Example 3 Structure 3
Behentrimonium Chloride	2.00	2.00	2.0	2.00
Cetearyl Alcohol	4.00	4.00	4	4.00
Xiameter MEM-7134	1.00	1.00	1.00	1.00
N,N,N-Trimethyl-2-octyldodecan-1-aminium chloride	—	0.38	—	—
2-(dioctylamino)-N,N,N-trimethyl-2-oxoethan-1-aminium	—	—	0.38	—
methanesulphonate N,N,N-trimethyl-2-((2-octyldodecyl)oxy)-2-oxoethan-1-aminium methanesulphonate	—	—	—	0.50
Parfum	0.60	0.6	0.6	0.60
Preservative	0.30	0.30	0.30	0.30
Aqua	to 100	to 100	to 100	to 100

The conditioners in examples A and 1 to 3 were prepared using the following method:

1. Surfactants and fatty materials (including branched materials) are added to a suitable vessel and heated to 80° C.
2. The molten blend is added to suitable amount of water according to the compositions disclosed, at a temperature between 45 and 70° C.
3. Mixture is mixed until opaque and thick.
4. The heat is then turned off, cooled to below 40° C., and the rest of the water is added along with the remaining materials.
5. Finally, the formulation is mixed at high shear on a Silverson mixer at 5000 rpm for 5 minutes.

Example 2: Treatment of Hair with Compositions a and 1-3

The hair used was dark brown European hair, in switches of 5 g weight and 6 inches in length.

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Hair was first treated with a cleansing shampoo using the following method:—

The hair fibres were held under running water for 30 seconds, shampoo applied at a dose of 0.1 ml of shampoo per 1 g of hair and rubbed into the hair for 30 seconds. Excess lather was removed by holding under running water for 30 seconds and the shampoo stage repeated. The hair was rinsed under running water for 1 minute.

The wet hair was then treated with the compositions using the following method:—

Conditioner was applied to the wet hair at a dose of 0.2 ml of conditioner per 1 g of hair and massaged into the hair for 1 minute. The hair was rinsed under running water for 1 minute and excess water removed.

Example 3: Silicone Deposition and Yield Stress onto Hair Treated with Compositions A and 1-3

TABLE 2

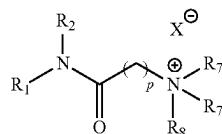
Yield stress and amount of silicone deposited on hair treated with Example A (comparative) and examples 1 to 4 (in accordance with the invention).

Ingredients	Example A Comparative	Example 1 Structure 1	Example 2 Structure 2	Example 3 Structure 3
Silicone Deposition [ppm]	431	1,565	1,185	1,354
Silicone Deposition ST DEV [ppm]	63	344	332	309
Yield stress [Pa] (±10 Pa)	182	87	55	167

The invention claimed is:

1. A composition comprising:

- (i) 0.01 to 10 wt % of a linear, cationic conditioning surfactant;
- (ii) 0.1 to 10 wt % of a linear fatty material;
- (iii) a particulate benefit agent selected from conditioning actives, scalp actives, encapsulated fragrance, emulsified fragrance, or mixtures thereof, wherein the particulate benefit agent includes at least silicone emulsions as the conditioning actives; and
- (iv) 0.01 to 5 wt %, at 100% active, of a branched cationic co-surfactant of the following structure



wherein:

- R₃ and R₄ comprise linear or branched alkyl chains, saturated or unsaturated, with carbon-carbon chain lengths of from C₆ to C₂₂;
- p has a range of from 1 to 6;
- R₇ comprises an alkyl chain having a carbon-carbon chain length of from C₁ to C₄;

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R₈ comprises a proton or an alkyl chain having a carbon-carbon chain length of from C₁ to C₄; and

X is an organic or inorganic anion;

wherein the molar ratios of branched cationic co-surfactants (iv) to linear cationic surfactants (i) are in the range of from 1:20 to 1:1.

2. The composition as according to claim 1, wherein the linear, cationic conditioning surfactant (i) has the formula 1: N⁺(R¹)(R²)(R³)(R⁴) wherein R¹, R², R³ and R⁴ are independently C₁ to C₃₀ alkyl or benzyl.

3. The composition according to claim 2, wherein the linear, cationic conditioning surfactant (i) is selected from behenyltrimethylammonium chloride, behenyltrimethylammonium methosulphate, cetyltrimethylammonium chloride, or mixtures thereof.

4. The composition according to claim 1, wherein the at least silicone emulsions do not comprise a hydrophobic modification.

5. The composition according to claim 1, wherein the at least silicone emulsions are selected from emulsions of dimethicone, dimethiconol, amodimethicone or mixtures thereof.

6. The composition according to claim 1, wherein the at least silicone emulsions are present in an amount of from 0.1 wt % to 10 wt % of the total composition.

7. The composition according to claim 1, wherein the branched cationic co-surfactant (iv) is present in an amount of from 0.1 to 2 wt %.

8. The composition according to claim 1, wherein the molar ratios of branched cationic co-surfactants (iv) to linear cationic surfactants (i) are in the range of from 1:10 to 1:1.

9. The composition according to claim 1, which has a yield stress in the range of from 30 to 200 Pascals (Pa) at 25° C. and 1 Hz.

10. The composition according to claim 1, wherein the X⁻ comprises an anion selected from a halide, a methanesulfonate group or an ethanesulfonate group.

11. A method of increasing deposition of a particulate benefit agent selected from conditioning actives, scalp actives, encapsulated fragrance, emulsified fragrance, or mixtures thereof, wherein the particulate benefit agent includes at least silicone emulsions as the conditioning actives, to hair comprising the steps of applying to hair the composition as defined in claim 1 and rinsing the hair with water.

12. The composition according to claim 1, wherein the at least silicone emulsions are present in an amount of from 0.25 wt % to 3 wt % of the total composition.

13. The composition according to claim 1, wherein the branched cationic co-surfactant (iv) is present in an amount of from 0.2 to 0.7 wt %.

14. The composition according to claim 1, wherein the molar ratios of branched cationic co-surfactants (iv) to linear cationic surfactants (i) are in the range of from 1:5 to 1:2.

15. The composition according to claim 1, which has a yield stress in the range of from 40 to 150 Pa peak value at 25° C. and 1 Hz.

16. The composition according to claim 1, wherein R₃ and R₄ comprise alkyl chains with carbon-carbon chain lengths of from C₆ to C₁₂.

17. The composition according to claim 1, wherein p has a range from 1 to 2.

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